



# POWER PIVOT & DATA MODELING

Build Relationships. Create Models. Get Powerful Insights.

## 1. INTRODUCTION

- Power Pivot is an Excel add-in that allows us to create **Data Models**, build relationships and perform **advanced analysis** on large datasets.
- It works behind the scenes using an in-memory engine called **VertiPaq**.
- Power Pivot is the foundation of DAX, Power BI and advanced reporting.

## 2. KEY BENEFITS

- ✓ Work with **Millions** of rows of data.
- ✓ Create **relationships** between multiple tables.
- ✓ Use **DAX** for powerful calculations.
- ✓ Build reusable data models.
- ✓ Improve performance with **VertiPaq Engine**.
- ✓ Perfect for **Dashboards, Reports** and Analysis.

## 4. HOW TO ENABLE POWER PIVOT

For Excel 2016 / 2019 / 2021 / Microsoft 365

- Go to **File** → Options → Add-ins.
- At the bottom, in Manage: **COM Add-ins** → Go.
- Check the box: **Microsoft Power Pivot for Excel**.
- Click **OK**.
- You will see the Power Pivot tab in Excel.

## 5. WHAT IS A DATA MODEL ?

- A **Data Model** is a collection of **tables**, **relationships** and **calculations** used to analyze data.
- It allows data from multiple sources to work together.
- Data Model is stored inside the **Excel** file.

## Traditional Excel vs Power Pivot

### Traditional Excel



- Row limit (1,048,576)
- Limited Relationships
- Slow with large data
- Hard to scale

### Power Pivot



- ✓ Millions of rows
- ✓ Multiple Relationships
- ✓ Fast In-Memory Engine
- ✓ Scalable & Flexible

## 3. REQUIREMENTS

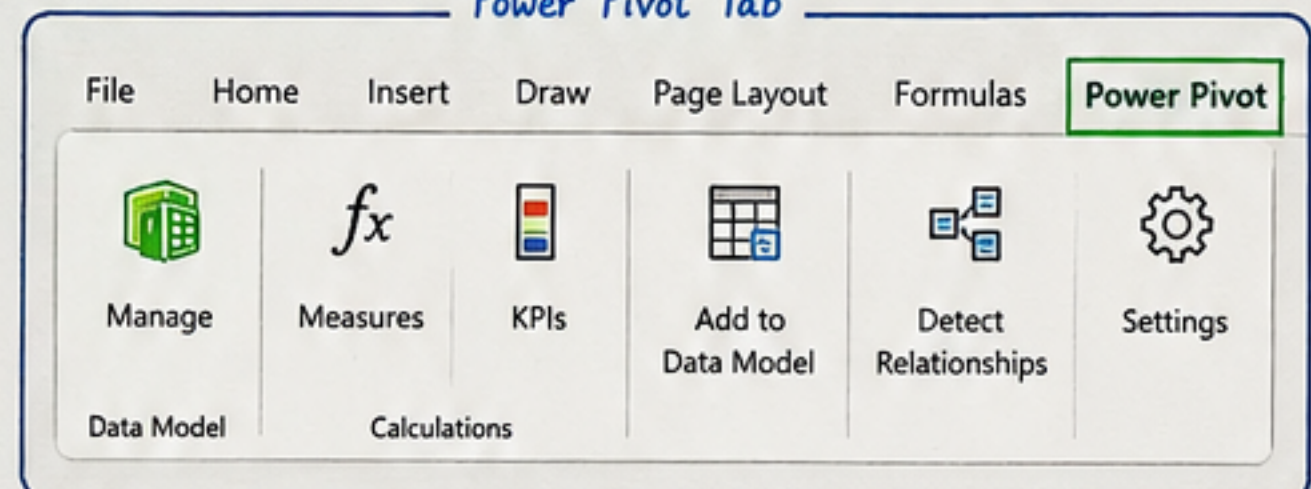
- Excel 2010 or later (recommended Excel 2016+)
- Windows Operating System
- At least 2 GB RAM (8 GB or more recommended)
- .NET Framework 3.5 or above
- 64-bit Excel (Recommended for large models)



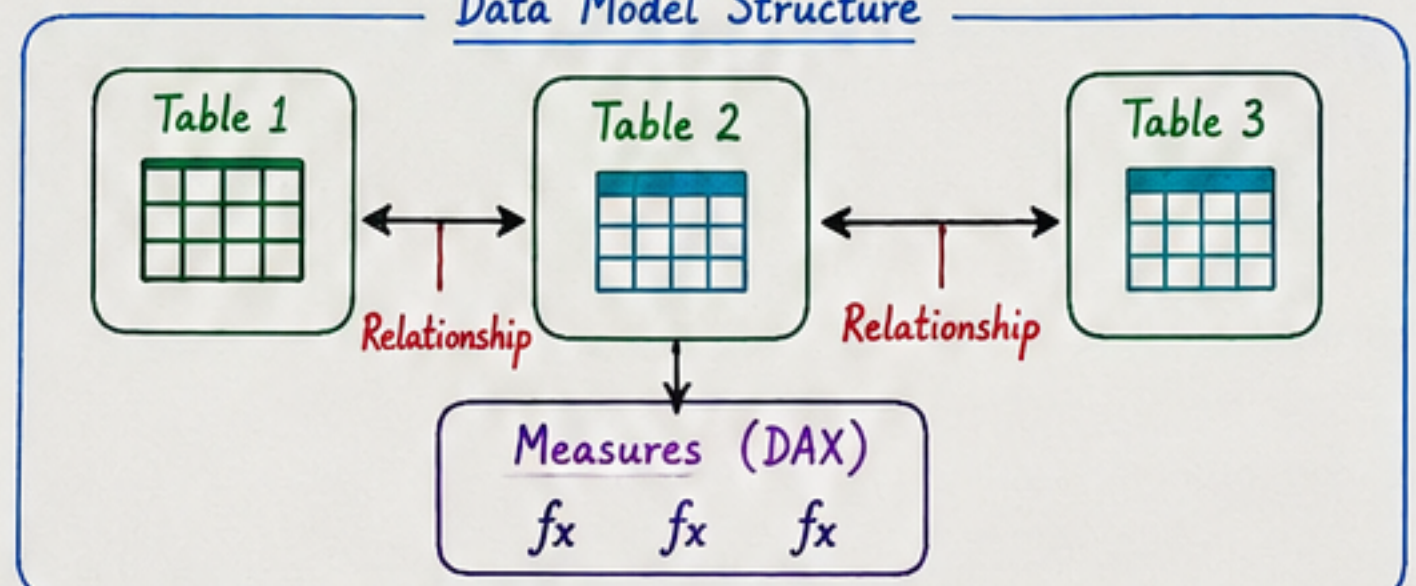
### Note:

Power Pivot is available only in Excel 2010 **Professional Plus** and higher versions.

### Power Pivot Tab



### Data Model Structure





## 6. DATA MODEL vs WORKSHEET DATA



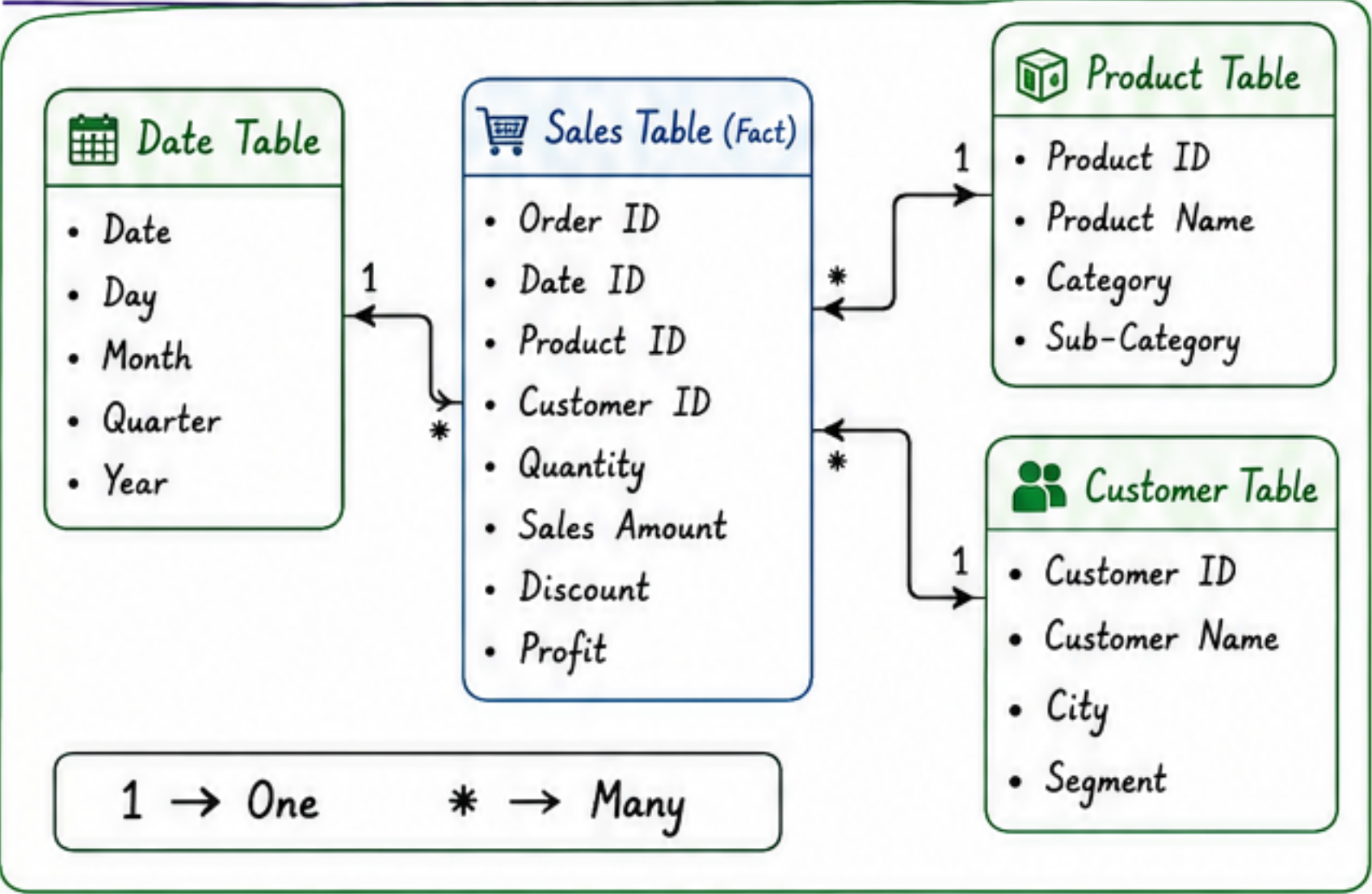
| WORKSHEET DATA                                                                                                                                                                                                                                                                                                                              | DATA MODEL (POWER PIVOT)                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Data stored in rows &amp; columns</li><li>• Each row is limited (1,048,576 rows)</li><li>• Difficult to create relationships</li><li>• Slower with large data</li><li>• Not ideal for advanced analysis</li></ul>  | <ul style="list-style-type: none"><li>• Data stored in VertiPaq engine</li><li>• Handles Millions of rows</li><li>• Easy to create relationships</li><li>• Very fast calculations</li><li>• Ideal for DAX, KPIs, Dashboards and Power BI</li></ul>  |



### Key Takeaway

Power Pivot turns **Excel** into a powerful **BI** tool by allowing us to store, relate and analyze large amounts of data.

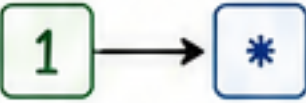
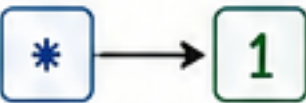
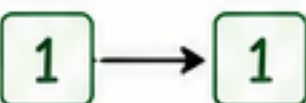
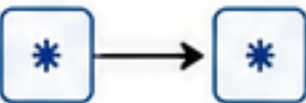
## 7. DATA MODEL STRUCTURE (EXAMPLE)



### Important Points

- Fact table contains **measures** (numbers).
- Dimension tables contain **descriptive** data.
- Relationships are created using **common keys**.
- **Filters** flow from Dimension tables to Fact table.
- This structure is called **Star Schema**.

## 8. TYPES OF RELATIONSHIPS

| Type               | Description                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| One to Many (1:*)  | The most common type. Used in Star Schema.                        |
| Many to One (*:1)  | Reverse of One to Many.                                           |
| One to One (1:1)   | Each value in one table matches only one value in another table.  |
| Many to Many (*:*) | Avoid if possible. Use bridge tables instead.                     |



### Best Practice

- Use **One-to-Many** relationships.
- Avoid **Many-to-Many**.
- Keep **Fact** table in the center.
- **Dimension** tables around the **Fact** table.



### Filter Direction

- Single Direction (Default): **Dimension** → **Fact**
- Both Direction (Use carefully): **Fact** ↔ **Dimension**

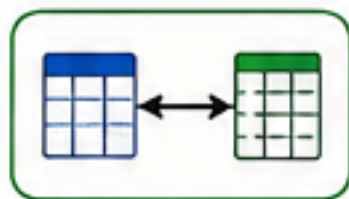


# 9. POWER PIVOT WINDOW OVERVIEW

Power Pivot window has 3 main views :

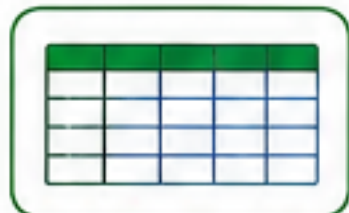
## ① Diagram View

Create relationships between tables and see the model visually.



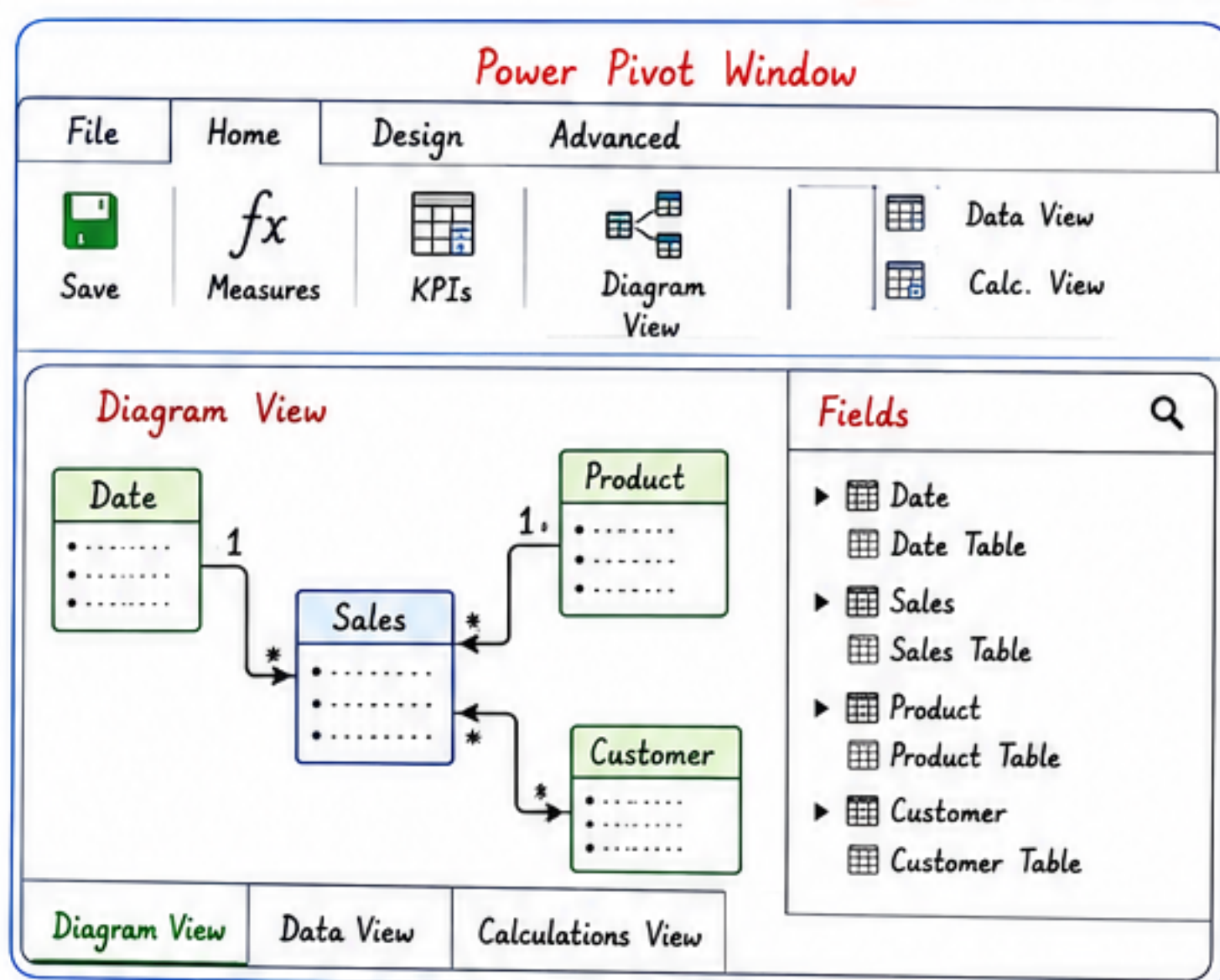
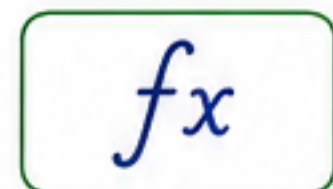
## ② Data View

See the data from all tables like a normal Excel sheet.



## ③ Calculations View

Create measures and calculated columns using DAX formulas.



# 10. MEASURES vs CALCULATED COLUMNS

| MEASURES                                                                                                                                                                                                                                                                            | CALCULATED COLUMNS                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Created using DAX.</li><li>Stored in memory.</li><li>Calculated on the fly.</li><li>Used in Pivot Tables, Charts, KPI's and Reports.</li><li>Performance is better.</li><li>Example: Total Sales, Profit, YoY Growth.</li></ul> <div>fx</div> | <ul style="list-style-type: none"><li>Created using DAX.</li><li>Stored in the table.</li><li>Calculated when data is loaded or refreshed.</li><li>Used as a new column in tables.</li><li>Performance is slower.</li><li>Example: Age, Category, Profit Margin %.</li></ul>  |



## When to Use What ?

- Use Measures for aggregations and calculations.
- Use Calculated Columns for row-level logic.
- Prefer Measures for better performance and memory efficiency.

# 11. DAX EXAMPLES (BASIC)

## ① Total Sales

Total Sales = SUM(Sales[Sales Amount])

## ② Total Quantity

Total Quantity = SUM(Sales[Quantity])

## ③ Total Profit

Total Profit = SUM(Sales[Profit])

## ④ Profit Margin %

Profit Margin % = DIVIDE([Total Profit], [Total Sales])

## ⑤ Distinct Customers

Distinct Customers = DISTINCTCOUNT(Sales[Customer ID])

## ⑥ Sales LY (Last Year)

Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))

# 12. BUSINESS USE CASES



**Sales Analysis:** Track sales, profit, region wise performance and trends.



**Customer Analysis:** Understand customer behavior, segments and lifetime value.



**Inventory Reports:** Monitor stock levels, slow moving items and inventory movement.



**Financial Reporting:** Create P&L, Balance Sheet and Cash Flow reports.

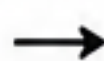


**Executive Dashboards:** Real-time KPIs, trends and key metrics in one view.

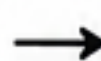
# 13. REAL LIFE EXAMPLE SCENARIO



1. Raw Data  
Multiple Excel Files



2. Power Query  
Clean, Transform & Load to Data Model



3. Data Model  
Create Relationships between tables



4. Reports  
Build Pivot Tables, Dashboards, KPIs

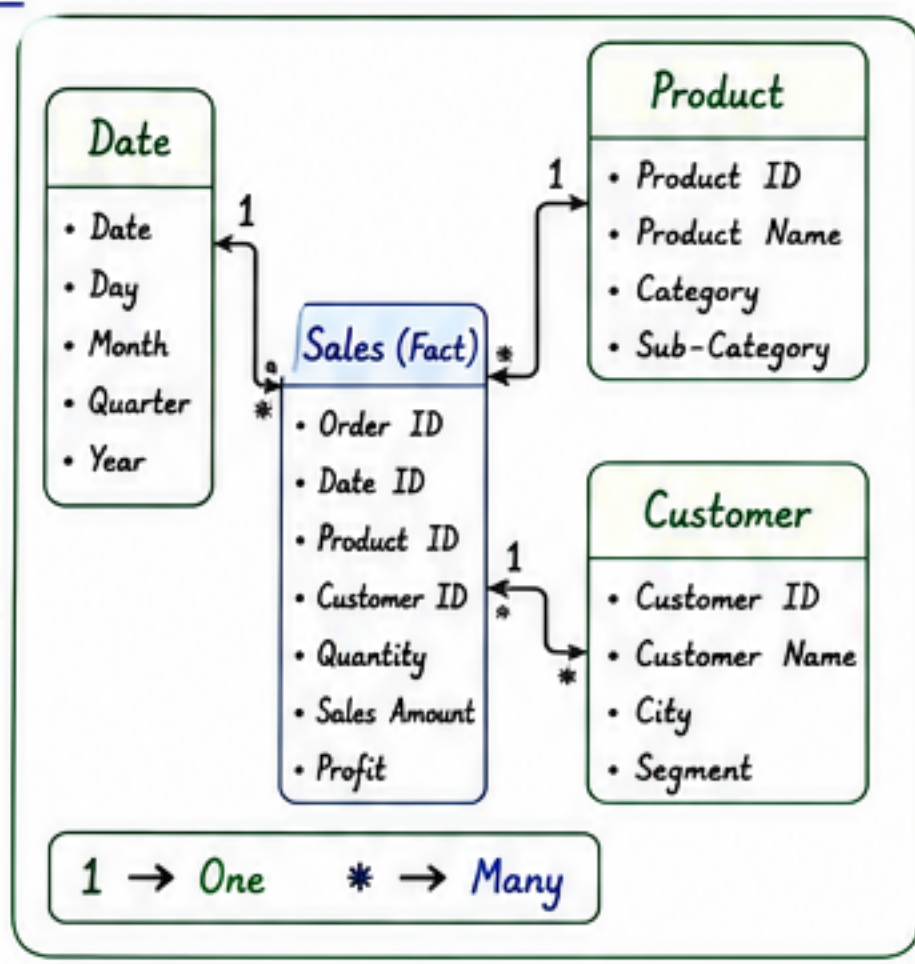
## Business Impact

Accurate Analysis • Faster Reporting • Better Decisions



14. COMMON POWER PIVOT FEATURES

- ★ **Measures** : Perform calculations using DAX.
- ★ **KPIs** : Track performance with targets.
- ★ **Calculated Columns** : Add new columns inside table.
- ★ **Relationships** : Connect tables using keys.
- ★ **Diagram View** : Visualize and manage the model.
- ★ **Data View** : See and edit data.
- ★ **Calculation View** : Create DAX measures and columns.





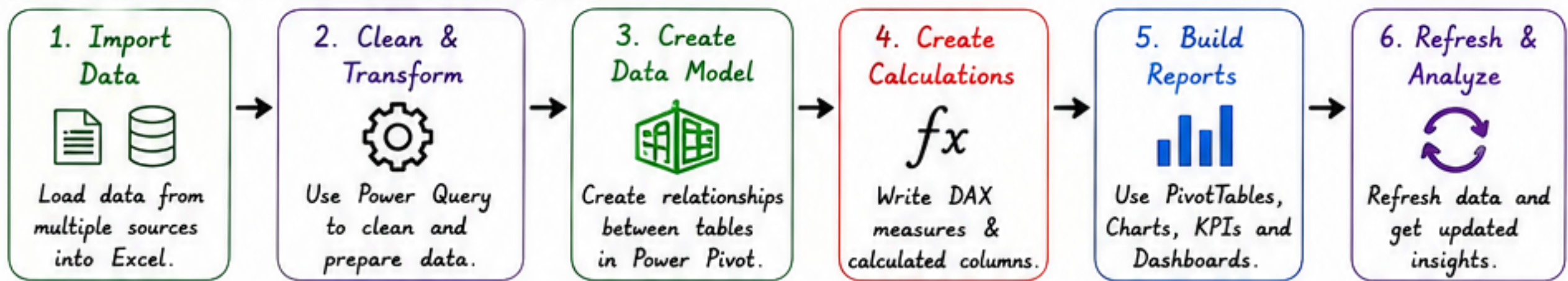
### Key Takeaway

Power Pivot helps us create a strong data model, perform lightning fast calculations using **DAX** and build powerful reports & dashboards in Excel or Power BI.

15. INTERVIEW QUESTIONS & ANSWERS

| Q.No. | Question                                                      | Answer (Short)                                                                                                                      | Key Point                     |
|-------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| ①     | What is Power Pivot?                                          | Power Pivot is an Excel add-in that allows us to create data models, build relationships and perform advanced analysis using DAX.   | Data Modeling & DAX Engine    |
| ②     | What is the difference between Excel and Power Pivot?         | Excel handles limited rows with limited relationships. Power Pivot handles millions of rows and multiple relationships efficiently. | Scale & Performance           |
| ③     | What is a Data Model?                                         | A Data Model is a collection of tables, relationships and calculations used to analyze data.                                        | Structure for Analysis        |
| ④     | What is the relationship between two tables?                  | A relationship connects tables using a common key column.                                                                           | One to Many (Most Common)     |
| ⑤     | What is a Fact table?                                         | A Fact table contains measurable numeric data like Sales, Profit, Quantity etc.                                                     | Numbers (Measures)            |
| ⑥     | What is a Dimension table?                                    | A Dimension table contains descriptive data like Product, Customer, Date, Region etc.                                               | Descriptions (Context)        |
| ⑦     | What is DAX?                                                  | DAX (Data Analysis Expressions) is a formula language used in Power Pivot for calculations.                                         | Powerful Calculation Language |
| ⑧     | What is the difference between Measure and Calculated Column? | Measure is calculated on the fly and stored in memory, Calculated Column is stored in the table.                                    | Performance & Storage         |
| ⑨     | Why use Power Pivot instead of VLOOKUP?                       | Power Pivot handles larger data, multiple relationships and is much faster for analysis.                                            | Scalability & Speed           |
| ⑩     | How does Power Pivot help in Dashboards?                      | It provides clean data model, fast calculations and feeds PivotTables, Charts, KPIs and Power BI dashboards.                        | Better Insights               |

16. POWER PIVOT WORKFLOW SUMMARY



17. BEST PRACTICES

- ✓ Follow **Star Schema** for your data model.
- ✓ Use meaningful names for tables & columns.
- ✓ Keep **Fact table** in the center.
- ✓ Use proper **data types** to improve performance.
- ✓ Use **Measures** instead of Calculated Columns whenever possible.
- ✓ Avoid unnecessary columns in your model.
- ✓ Test your **DAX** formulas step by step.

18. PRACTICAL BUSINESS USE CASES

**Sales Performance Dashboard**

**Customer Segmentation Analysis**

**Inventory & Stock Analysis**

**Financial Reporting & P&L Analysis**

**KPI Tracking & Goal Analysis**

Power Pivot turns raw data into meaningful business insights!

19. REMEMBER

- ★ Power Pivot = Data Modeling + DAX
- ★ Build relationships, not lookups.
- ★ Use Measures for calculations.
- ★ Design once, use everywhere.
- ★ Clean Model = Accurate Insights.

**MASTER DATA MODELING.**  
UNLOCK THE POWER OF DAX.  
BUILD INSIGHTS THAT MATTER.